S. S. GOKULA SWAMY

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Research & Achievements

IEEE AIC 2026 (Accepted Research Paper) Bengaluru, India
Co-Author — GPU-Accelerated Hybrid GA for Wind Farm Optimization (wflo-ga) *2026*

- **GPU Wake Vectorization**: Co-authored accepted paper optimizing 150 turbines on irregular polygons (10 km × 10 km); vectorized Jensen wake model in PyTorch ((8, 150, 150) tensors, 8 wind directions) eliminating Python loops during screening.
- **Dual-Fidelity GA**: Validated elite layouts via conditional NREL FLORIS rescoring with a 0.88 fitness gate and 10-call per-generation budget cap; achieved **4029.16 GWh/yr** Annual Energy Production (**+2.69% / +105.7 GWh/yr AEP gain** over baseline, outperforming TOPFARM and Jensen-only GA) while satisfying ≥ 400 m turbine spacing via tiered repair operators.
- **Tech Stack**: PyTorch (CUDA Tensors), FLORIS (NREL), PyGAD, Shapely, TopFarm, PyWake, NumPy.

UNCLONE AI Hackathon 2026 (BMSCE & DetectifAI) Bengaluru, India
1st Runner-Up — Verified-Agent Badge (Team 39) *Oct 2026*

- **Real-Time Voice Deepfake Defense**: Built a client-side agent verifying banking representative authenticity in real time over live calls in 6-second windows; combined DetectifAI speaker verification with a segment-level policy where synthetic confidence ≥ 0.90 on any 4-second chunk triggers an instant transaction hold, overriding reassuring call averages.
- **Zero-Dependency Signal Forensics**: Authored a pure Python stdlib PCM WAV forensics module detecting anomalies (silence tell > 0.15, room RMS < 0.004, clipping); implemented in-session Web Audio pitch/spectral voice conversion for discrimination testing.
- **Tech Stack**: FastAPI, Python (stdlib Audio Forensics), DetectifAI API, Web Audio API, Uvicorn, Render.

Education

B.M.S. College of Engineering Bengaluru, India
B.E. in Artificial Intelligence & Machine Learning | ***CGPA: 7.37 / 10*** *(7th Semester)* *2023 – 2027*

- **Major Project**: *NutriSense* — Autonomous 3D dietary assessment and statutory PM POSHAN compliance auditing.
- **Relevant Coursework**: Large Language Models, Deep Learning, Time Series Analysis, Evolutionary Computing, Graph ML.

Sainik School Kodagu Kodagu, India
CBSE Class 12: 88.6% (2023) | *CBSE Class 10: 91.4% (2021)* *2016 – 2023*

- **Honors & Leadership**: Qualified NDA Written Exam (2023, top ∼0.5% nationwide); Best Volunteer, NSS Annual Camp 2025 (selected from 100+ participants); completed 79.9 km and 58.2 km endurance cyclothons.
- **Military & Cadet Credentials**: Earned **NCC Certificates 'A' (Grade A), 'B' (Grade B), and 'C' (Grade B)**.

Projects

NutriSense | *SigLIP-2, DINOv2-S, MoGe-2, SAM 2.1, PyTorch, React 19* *github.com/Lion4467cat/NutriSense*

- **Multimodal Perception & Mass Fusion**: Architected an 8-subsystem single-photo 3D dietary metrology pipeline for PM POSHAN Mid-Day Meals; deployed Google SigLIP-2 (768-dim) with $\alpha = 0.65$ background attenuation classifying 115 regional dishes without food bleed; reconstructed 3D volume using MoGe-2 metric depth with SVD tabletop plane fitting (3.5 mm tare) and conical frustums.
- **Inverse-Variance Fusion & Statutory Audit**: Trained frozen Meta DINOv2-S weight head with SmoothL1 loss, fused with geometry via inverse-variance weighting ($w_{\text{geom}} \approx 0.27$, $w_{\text{head}} \approx 0.73$); audited IFCT 2017 nutrient bounds via a 4,000-draw Monte Carlo engine applying Fréchet–Hoeffding comonotonic sums; served via FastAPI with CUDA OOM guards to React 19.

The Greenifier | *OpenCV, Flask, NumPy, NASA POWER, Llama 3.3 70B* *github.com/Lion4467cat/greenifier*

- **Geospatial Computer Vision**: Built a rooftop solar estimator fetching keyless Zoom 18 satellite tiles via Web Mercator projection; segmented usable roofs and terrain via OpenCV HSV thresholding, Canny edges, and morphological closing.
- **LLM-Driven Energy Auditing**: Integrated NASA POWER API for location solar irradiance and Llama 3.3 70B (Groq) for financial modeling (system size, INR 65,000/kW capex, payback years, CO_2 offset) via a responsive Flask and Leaflet.js dashboard.

Technical Skills

- **AI Systems & Multimodal Pipelines**: Google SigLIP-2, Meta DINOv2, Microsoft MoGe-2, Meta SAM 2.1, LLM Orchestration, Spec-Driven Development, Evolutionary Computing, Monte Carlo Uncertainty Modeling, 3D Metric Depth.
- **Languages**: Python, SQL, HTML5, CSS3, JavaScript (ES6+).
- **Frameworks & Libraries**: PyTorch, FastAPI, Flask, React 19, Leaflet.js, OpenCV, NumPy, Shapely, PyGAD, TopFarm, FLORIS, Web Audio API.
- **Developer Tools & Platforms**: Git, GitHub, Linux, Uvicorn, Vite, uv, Docker, Render, Groq API, NASA POWER API, DetectifAI API.